

1.2 Applications of HCI (Domains of Life)

ایچ سی آئی کی ایپلیکیشنز صحت، بینکنگ، تعلیم اور نیٹ ورکنگ کے شعبوں میں انسانی ضروریات پوری کرتی ہیں۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 03 of 15

★ Golden

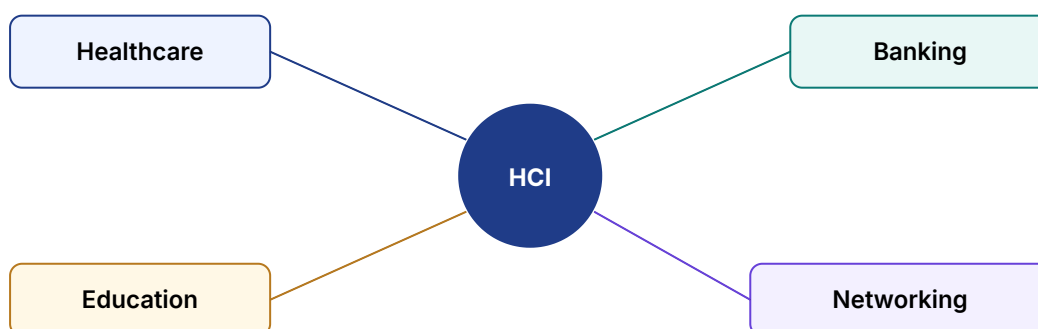
1.2 Applications of HCI (Domains of Life) ★ Golden

LEARN IT

HCI is used to design and develop interactive systems such as **desktop applications, websites and mobile apps**. These are **human-centred solutions** built to meet user needs across many domains of life. A domain is an area of daily activity — health, money, school, talking to people.

When you design for a domain, you design for that domain's **users, tasks and risks**. A mistake on a games app is annoying. A mistake on a medical or banking screen can harm a person or lose money. That is why examiners treat this topic as golden.

Domain	Application examples	User benefit
Health care	Patient portals, telemedicine apps, assistive technologies	Improves access to medical services remotely
Banking	Mobile banking apps, secure ATM interfaces	Enables quick transactions and secure financial management
Education	E-learning platforms (LMS), online exam portals	Enhances teaching, learning and student engagement
Networking and communication	Social media, messaging apps (WhatsApp), Zoom	Connects people globally through collaboration tools



HCI sits at the centre of four everyday domains: healthcare, banking, education and networking.

WORKED EXAMPLE

Worked example: the same HCI idea in three Pakistani systems.

Step 1 — ATM (banking). You insert a card, the screen shows large buttons (Fast Cash, Balance, Other), you enter a PIN on the keypad, and you take cash. HCI benefit: a quick, secure transaction without a human teller.

Step 2 — JazzCash (banking + communication). You log in with a PIN or fingerprint, choose Send Money, type a number and amount, confirm, and see "Successful" with a JazzCash SMS. HCI benefit: money moves in seconds from a shop in Larkana to a student in Karachi.

Step 3 — School portal (education). A Grade XII student opens the college LMS, taps "Download date sheet", and submits a leave form. HCI benefit: learning and admin tasks happen without standing in a queue at the office.

Step 4 — Compare the user benefit. Banking HCI saves time and keeps money safe. Education HCI raises engagement. Health HCI (a Sehat/telemedicine app) brings a doctor to a village phone. Networking HCI (WhatsApp, Zoom) keeps a family in Dubai in touch with parents in Sukkur.

REMEMBER IT

Exam mnemonic H-BEN: Health, Banking, Education, Networking. For each, write: one application + one user benefit. That structure scores full marks on "applications of HCI".

COMMON MISTAKE

Mistake: Listing brand names only ("WhatsApp, JazzCash, Zoom") without saying the *domain* and the *user benefit*.

Fix: Always write three parts: domain → example → what the user gains. "Banking: JazzCash mobile app — enables quick, secure transfers."

اردو میں سمجھیں

ایچ سی آئی کی ایپلیکیشنز صحت، بینکنگ، تعلیم اور نیٹ ورکنگ کے شعبوں میں انسانی ضروریات پوری کرتی ہیں۔ صحت میں مریض پورٹل اور ٹیلی میڈیسن دور سے علاج دیتے ہیں، بینکنگ میں موبائل ایپ اور اے ٹی ایم تیز اور محفوظ لین دین دیتے ہیں، تعلیم میں ای لرننگ اور آن لائن امتحان مشغولیت بڑھاتے ہیں، اور واٹس ایپ یا زوم لوگوں کو جوڑتے ہیں۔

EXAM TIP

Very common 5–8 mark question: "Explain applications of HCI in different domains of life with examples." Draw a four-row table (Health care, Banking, Education, Networking & communication). Each row: one or two application names + the user benefit from the source. Add a local line (ATM / JazzCash / school portal) to show you can apply the idea. Do not write a story about only one app.

CHECK YOURSELF

1. Name the four domains of HCI given in the source.
2. What user benefit does HCI bring in banking?
3. Give one education application and one networking application.

Answers: 1. Health care, banking, education, networking and communication. 2. Quick transactions and secure financial management. 3. Education: LMS or online exam portal. Networking: social media, WhatsApp or Zoom.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. HCI applies to:

- a. Desktops, web, mobile, kiosks, embedded, VR
- b. Only mainframes in 1970
- c. Only NAND trainers
- d. Only paper

MCQ 2. A hospital infusion pump UI is HCI in:

- a. Health / safety-critical life
- b. Sports memes only
- c. Fashion only
- d. OSI cabling only

MCQ 3. Education HCI example:

- a. LMS that students can actually submit homework on
- b. A random 4pt PDF
- c. An unlabelled form
- d. A hidden button

MCQ 4. Bad HCI in a bank app can:

- a. Make people send money to the wrong account
- b. Speed bubble sort
- c. Fix NAND
- d. Change Big O of Python

MCQ 5. Domains of life in the book include:

- a. Work, health, education, home, public services
- b. Only gaming
- c. Only compilers
- d. Only ER

MCQ 6. This is ★ because you must:

- a. Name domains with a concrete interface each
- b. Draw K-maps
- c. Write SQL only
- d. List sprints

Extra short questions

1. Five domains of HCI application.
2. Why HCI in medical devices is life-critical.
3. A public-service example in Pakistan.
4. What "application of HCI" means in a 3-mark.

Extra long question

1. Pick four domains. For each: one system, one user, one HCI requirement. ★ Golden 10-mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Work software, health devices, education, home/consumer, government/public kiosks, transport...
2. A confusing unit (mg vs ml) can dose wrongly.
3. NADRA/passport kiosks, metro ticket machines.
4. Not the definition — a place + the interface + why users matter.

Long

1. Clinic: nurse, glanceable alarms. College portal: student, language + mobile. ATM: elderly, privacy hood + large text. Car dash: driver, eyes-up. Table.

Next lecture: 1.3 Components of HCI